Assignment - 1

Q1

a) i) $\frac{108-100}{100} = 0.08$ 8% Return

ii) $\frac{103-108}{108} = \frac{-5}{108} = -0.0463$

-4.63% Return

iii) $\frac{115-103}{103} = 0.1165$

11.65% Returns

iv) $\frac{110-115}{115} = -0.0435$

-4.35% Returns

b) Log Returns

i) $\log(108/100) = 0.0770$

ii) $\log(103/108) = -0.474$

iii) $\log(115/103) = 0.1102$

iv) $\log(110/115) = -0.0445$

c)

Return	Log Return	Difference
0.08	0.0770	0.0030
-0.0463	-0.474	0.0011
0.1165	0.1102	0.0063
-0.0435	-0.0445	0.0010

Max Difference for 3rd Month

Q2)

a) $R_p = 0.4 R_1 + 0.6 R_2$

$E(R_p) = E(0.4R_1 + 0.6R_2)$

$= E(0.4R_1) + E(0.6R_2)$

$\boxed{= 0.4\, E(R_1) + 0.6\, E(R_2)}$

$0.4 \times 0.15 + 0.6 \times 0.07$

$0.060 + 0.042$

$= 0.102$

$= 10.2\%$

b) No, Expected Return will the same as no term of corelation is present

Q3)

$Var(X+Y) = Var(X) + Var(Y) + 2\,Cov(X,Y)$

Let

$R_p = w_1 R_1 + w_2 R_2$, $Var(R_1) = \sigma_1^2$

$Var(R_2) = \sigma_2^2$

$\frac{Cov(R_1,R_2)}{\sigma_1 \sigma_2} = CoRelation(R_1, R_2)$

$\frac{Cov(R_1,R_2)}{\sigma_1 \sigma_2} = \rho_{1,2}$

So

$Var(R_p) = Var(w_1 R_1 + w_2 R_2)$

$= Var(w_1 R_1) + Var(w_2 R_2) + 2Cov(w_1 R_1, w_2 R_2)$

$= w_1^2 Var(R_1) + w_2^2 Var(R_2) + 2w_1 w_2 Cov(R_1, R_2)$

$= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 Cov(R_1, R_2)$

$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$

Q4)

	Returns	stan Deviation	weights
A:	$\mu_A = 0.18$	$\sigma_A = 0.25$	$w_A = 0.5$
B:	$\mu_B = 0.08$	$\sigma_B = 0.12$	$w_B = 0.5$

a) from last Question,

$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$

$\sigma_p^2 = 0.25(0.25)^2 + 0.25(0.12)^2 + 0.5(0.25)(0.12)\rho$

$\sigma_p^2 = 0.015625 + 0.0036 + 0.015\rho$

i) $\rho = +1$ $\sigma_p^2 = 0.034225$

$\sigma_p = \sqrt{0.034225} = 0.185$

ii) $\rho = 0$ $\sigma_p^2 = 0.019225$

$\sigma_p = \sqrt{0.019225} = 0.13865$

iii) $\rho = -1$ $\sigma_p^2 = 0.004225$

$\sigma_p = \sqrt{0.004225} = 0.065$

b) We know

$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$

if $\rho = 1$, $\sigma_p^2 = \underbrace{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2}_{(w_1\sigma_1 + w_2\sigma_2)^2}$

$\sigma_p^2 = (w_1\sigma_1 + w_2\sigma_2)^2$

$\sigma_p = w_1\sigma_1 + w_2\sigma_2$

Weighted Avg of individual Risks when $\rho = +1$

c) $\rho = -1$ so, $\sigma_p^2 = (w_1\sigma_1 - w_2\sigma_2)^2$

$\sigma_p = w_1\sigma_1 - w_2\sigma_2$

let $w_1 = x$ so $w_2 = 1-x$

$\sigma_p = x\sigma_1 - (1-x)\sigma_2$

$0 = x \cdot 0.25 - 0.12 + x \cdot 0.12$

$0.12 = 0.37x$

$x = \frac{12}{37}$

$x = 0.3243$

Q5)

$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$

if $\rho = 1$, $\sigma_p^2 = \underbrace{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2}_{(w_1\sigma_1 + w_2\sigma_2)^2}$

so $\sigma_p = w_1\sigma_1 + w_2\sigma_2$

if $\rho < 1$, then $2w_1 w_2 \sigma_1 \sigma_2 \rho < 2w_1 w_2 \sigma_1 \sigma_2$

then overall term $< (w_1\sigma_1 + w_2\sigma_2)^2$

Hence $\sqrt{\text{overal term}} < w_1\sigma_1 + w_2\sigma_2$

Hence σ_p will be lower for $\rho < 1$ than $\rho = 1$

Q6)

Asset	μ_i	σ_i
1	15%	25%
2	8%	12%
3	5%	4%

Corelation matrix ρ

$$\begin{array}{c} 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 1 & 0.4 & 0.1 \\ 0.4 & 1 & 0.2 \\ 0.1 & 0.2 & 1 \end{bmatrix}$$

a) $\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$

$\Sigma_{11} = \sigma_1^2 = 0.0625$

$\Sigma_{22} = \sigma_2^2 = 0.0144$

$\Sigma_{33} = \sigma_3^2 = 0.0016$

$\Sigma_{12} = \Sigma_{21} = \rho_{12} \sigma_1 \sigma_2 = 0.4 \times 0.25 \times 0.12 = 0.012$

$\Sigma_{13} = \Sigma_{31} = \rho_{13} \sigma_1 \sigma_3 = 0.1 \times 0.25 \times 0.04 = 0.001$

$\Sigma_{32} = \Sigma_{23} = \rho_{23} \sigma_2 \sigma_3 = 0.2 \times 0.12 \times 0.04 = 0.00096$

$$\begin{bmatrix} 0.0625 & 0.012 & 0.001 \\ 0.012 & 0.0144 & 0.00096 \\ 0.001 & 0.00096 & 0.0016 \end{bmatrix}$$

b) as $\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$

so $\Sigma_{ji} = \rho_{ji} \sigma_j \sigma_i$

we know $\rho_{ij} = \rho_{ji}$

so $\Sigma_{ij} = \Sigma_{ji}$

hence Symmetry

Q7)

a) X: $\frac{14-4}{22} = \frac{10}{22} = 0.4545$

Y: $\frac{9-4}{10} = \frac{5}{10} = 0.5$

Z: $\frac{20-4}{35} = \frac{16}{35} \approx 0.4571$

b) Sharpe Ratio $= \frac{E(R_p) - R_f}{\sigma_p}$] Return] Risk

Investor wonts max Return with minimum Risk, so the Portfolio with max Sharpe Ratio is the best

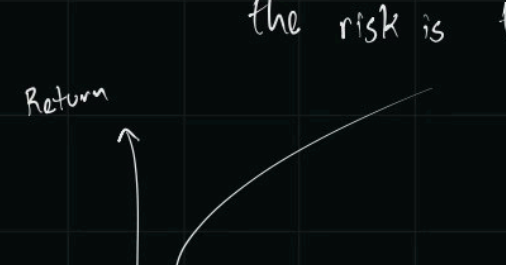
so $Y > Z > X$

c) Z has highest Return, But Also Highest Risk. Return per unit Risk is higher for Y.

Hence Prefer Y.

Q8)

a) GVM: A portfolio of Assets with weights adjusted such that the risk is the minimum.



As GVM have least risk, ie least standard deviation, so it must have least Variance

Hence Left most point.

b) It is Incorrect.

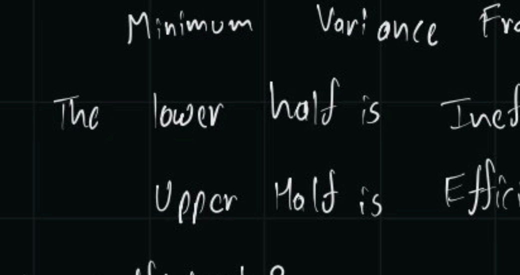
Minimum Variance Frontier has 2 portion, Efficient & Inefficient.

The lower half is Ineffecient

Upper Half is Efficient

why Inefficient?

lower half offer lower Returns than Upper half at the exact same Risk.



More Return at Same Risk.

c) By introducing a safe Asset like gov. bonds with a Risk free fixed Return.

Y intercept = Risk free Rate of Return

Slope: Make slope such that it perfectly tangents the Efficient Variance Frontier.

slope of CML Represents sharpe Ratio of the Market.

Q9)

a) $(Error)^2 = \frac{\sigma_i^2}{T}$ Assuming σ_i was yearly volitility

$Error = \frac{\sigma_i}{\sqrt{T}} = \frac{0.30}{\sqrt{3}} \approx 0.1732$

Historical Data is Crucial to find the Confidence intervals to Assess how good the Investment Really is.

b) MVO assumes the μ and σ provided to it are absolute. It dosent Account how confident we are About the data.

As in part a, we calculatet the error is 17.3% even after 3years of data.

ML solves this with Regularization the data base on the Historical data and values.

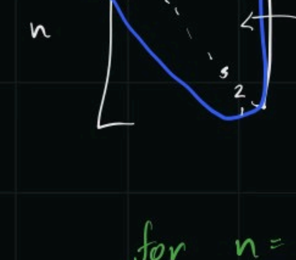
c) Unique Parameters in $\Sigma = \overbrace{{}^nC_2}^{\text{ways to pick 2 diff Assets}} + \overbrace{n}^{\text{for diagonal ones (same 2 assets)}}$

$\frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2} + n$

$= \frac{n^2 - n + 2n}{2}$

$= \frac{n^2 + n}{n} = \frac{n(n+1)}{2}$

or



element in upper triangle of Σ

ie sum of $1 \to n$

$\frac{n(n+1)}{2}$

for $n = 50$,

$\frac{50(51)}{2} = 1275$ parameters

to calculate this many parameters, we must have a large Amount of Data to accurately calculate Each parameter.